

Design of Nonverbal Symbolic Visual Cues for VR Social Interaction Based on AI Semantic Recognition

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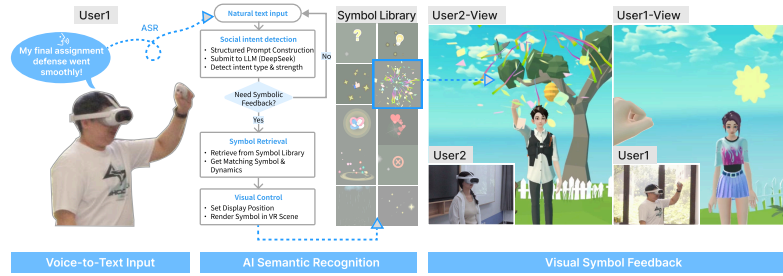


Figure 1: User experience flow of the AI-driven nonverbal symbolic visual feedback system.

Abstract

This study introduces a nonverbal symbolic visual feedback system driven by AI semantic recognition, aimed at enhancing emotional expression and interaction understanding in virtual reality social environments. We designed visual symbols for ten representative social behaviors. The system can recognize the type and intensity of social intent from users' natural speech in real time and automatically trigger corresponding 3D visual feedback. Experimental results show that the system significantly improves the user experience in remote social interaction.

1 Introduction

As virtual reality technology advances, VR-based social interaction is redefining how people connect remotely. While users can interact through avatars in 3D spaces, the limited availability of nonverbal cues still hinders accurate emotional expression and social understanding. Recent efforts in social VR have enhanced avatars with body gestures and facial expressions to simulate real-world emotional exchange, but these approaches often rely on high-precision sensors, which suffer from accuracy issues, latency, and poor synchronization with real movements—making them unsuitable for lightweight everyday interactions[Chen et al. 2020].

Research in computer-mediated communication (CMC) has shown that nonverbal visual elements, such as typography and emojis, can effectively enrich emotional expression in remote textual communication. Some studies have begun to explore the integration of emojis[Shen et al. 2024] and visual particles[Lee et al. 2022] into VR social contexts to enhance interaction. However, these visual cues typically require manual triggering and lack contextual semantic

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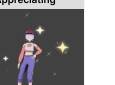
Social Behavior	Questioning	Agreeing	Consoling	Encouraging	Sympathizing
VR Social Symbols					
Spatial Dynamic	Above the head Slowly rising and rising	On the hand Bouncing, flying toward the other	In front of chest Slowly emerging	Surrounding Rhythmic bouncing	Above the head Sinking and rising
Social Behavior	Wishing	Sharing opinion	Expressing care	Disapproving	Appreciating
VR Social Symbols					
Spatial Dynamic	Above the avatar Exploding	Above the head Slowly rising	Near the chest Growing larger and flying toward the other	Above the head Faint twinkling	Surrounding Twinkling

Figure 2: Design of Nonverbal Visual Symbols in VR for Ten Types of Social Behaviors

awareness, limiting their integration into natural, speech-based dialogue.

To address this, we propose an AI semantic recognition-driven system for nonverbal, symbol-based visual feedback in VR. The system detects users' social intentions from spoken dialogue in real time and automatically triggers corresponding 3D symbolic visuals (e.g., celebratory ribbons), leveraging spatial and dynamic expression to enhance the visibility and clarity of emotional cues—thereby improving immersion and social understanding in VR environments.

2 Design of Social Symbols in VR

In this system, when users engage in social conversations within a virtual reality environment, the AI semantic recognition module analyzes their spoken content in real time to detect social intentions and automatically triggers matching nonverbal visual symbols. These symbols dynamically appear around the user's avatar.

2.1 Design of Social Symbols in VR

Social Behavior Classification. In designing the system, we referenced the social intent taxonomy proposed by Welivita et al. [Welivita and Pu 2023] and selected ten representative social behavior types suitable for visual expression. We included a "no feedback" category to avoid redundant or distracting outputs.

Spatial and dynamic design of social symbols. Based on the ten identified social behaviors, we designed a set of semantically mapped 3D symbols for each. Spatially, the symbols emerge from specific parts of the avatar's body (e.g., hearts float from the chest), creating embodied channels of expression. Dynamically, all symbols feature motion effects such as bouncing and trailing, enhancing emotional intensity and expressive richness during interaction.

Design Based on Social Intent Strength. Each type of social behavior is divided into three levels of intent strength: weak, medium, and strong. The system identifies the expressed intent level in real time and dynamically adjusts the visual symbol parameters accordingly, including quantity, size, and color.

2.2 AI Recognition Module

We implement a low-latency pipeline that links speech, text, semantic inference, and visualization. Audio is transcribed with streaming ASR into punctuation-preserving clauses; for each clause, a sliding-window dialogue context (recent turns, speaker tags, timestamps,

last prediction, revocations) is encoded as a schema-constrained JSON prompt to a large language model. The model returns a social behavior label from a ten-class taxonomy with an explicit *no-feedback* option and a three-level intent strength (weak/medium/strong) under deterministic decoding. To balance stability and responsiveness, we apply a confidence check, simple debouncing, and light temporal smoothing, abstaining on uncertainty. Accepted predictions drive symbol retrieval and parameterization (quantity, scale, lifetime, emission rate), while scene-aware constraints prevent occlusion and preserve legibility; safety rules route inappropriate or underspecified content to *no-feedback*, and users may revoke the last emission via controller input.

3 Conclusion and Future Works

We conducted a comparative study with four participant groups (4 female, 4 male) using the simplified User Experience Questionnaire (UEQ-S). Results showed that adding nonverbal visual feedback significantly improved user experience: pragmatic quality rose from -0.250 to 1.594, hedonic quality from -1.656 to 2.531, and overall experience from -0.953 to 2.063. This confirms our approach effectively enhances communication quality and immersion in VR social interaction. The integrated visual feedback strengthened emotional expression, clarified semantics, and reduced detachment and misunderstanding in virtual contexts. The system shows strong potential for remote companionship, online meetings, and virtual concerts.

Future work will incorporate prosodic and tonal cues for more accurate intent detection, explore auditory feedback to reinforce social presence, and conduct larger IRB-approved studies across diverse social scenarios to assess adaptability and scalability.

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